

Global Political Power Shifting: from Oil to Renewable Energy

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I. Introduction

Oil holds immense importance as a vital and highly valuable natural resource on a global scale. It has played a significant role in shaping economies, exerting influence on geopolitical dynamics, and serving as a key energy source for diverse industries. The importance of oil stems from its versatility as an energy source and its extensive use in transportation, manufacturing, and numerous other sectors. Beyond its industrial applications, oil has evolved into a source of political power, exerting a profound influence on the trajectory and destiny of nations. Nevertheless, the dependency on oil presents inherent challenges and risks such as volatility in oil prices that can profoundly impact the global economy, influencing inflation rates, trade balances, and overall economic stability. Furthermore, the extraction, production, and consumption of oil contribute to environmental consequences, most notably climate change.

As a response, renewable energy has emerged as a vital and transformative solution to tackle the challenges posed by climate change, greenhouse gas emissions, and energy security, which cannot be adequately addressed by oil. Unlike oil, renewable energy sources offer a more sustainable and secure energy supply, mitigating the risks associated with price fluctuations and geopolitical tensions. These fundamental changes in the global energy system will have far-reaching geopolitical consequences and will affect nearly all nations since they have grown quicker than any other energy source due to technological advances and decreasing costs.

The rapid adoption of renewable energy will reshape the distribution of political power and international relations among nations, similar to how fossil fuels have historically shaped the geopolitical landscape and the energy transformation. Therefore, the objective of this study is to examine the impact of renewable energy growth on shifting global political power and influence between countries. This study will employ data analysis to quantify the political power of oil, renewable energy, and overall power. Through correlation analysis, we aim to provide evidence supporting the hypothesis that the transition from oil to renewable energy has the potential to reshape global political power relations. This study provides valuable insights into global energy trends to private energy stakeholders and individuals interested in energy policy. Furthermore, it can serve as a resource for students seeking to deepen their understanding of the evolving dynamics in the field of energy policy.

This study is divided as follows, section 1 Introduction, section 2 will cover the dominance of oil in global politics and analyze the significance of oil in geopolitics. In section 3, the environmental and economic challenges of oil will be looked into, and the rise of renewable energy will be covered in section 4 to see the renewable energy sources, potential, and which country will be the leader. Section 5 will delve into the implications of transitioning from oil to renewable energy, examining the transformations occurring in the global energy landscape and the subsequent impact on geopolitical alliances and rivalries. Lastly, section 6 will provide a focused conclusion summarizing the key findings and insights of the study.

II. Historical significance of oil in geopolitics

The significance of oil as the primary energy source gained prominence during the 1960s and 1970s. By 1950, oil's share of global energy production was less than one-third, but within two decades, it surpassed coal, which was the predominant commercial energy source at that time, accounting for 75% of global energy. This shift from coal to oil was justified by its cleaner nature, higher efficiency, and lower cost. Consequently, oil became the primary driver of industrialization in the Global North, increasing at the same time the dependence on oil revenues from the Global South (Euclid R., 2004). After half a century, oil continues to dominate as the principal energy source, as is shown in Figure 1, which has involved several political implications and power dynamics within this sector.

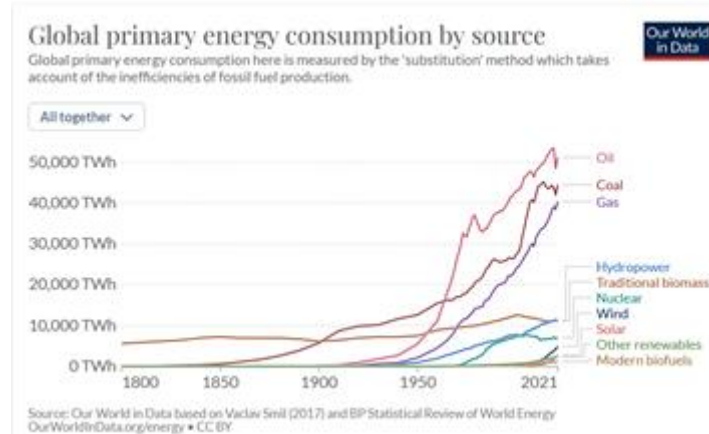


Figure 1. Global primary energy consumption by source

Source: Our World in Data, 2021

The discovery of oil reserves in the Middle East in the early 20th century, for instance, made countries like Saudi Arabia, Iran, Iraq, and Kuwait important players in world oil production, transforming their economies and political influence. However, initially, many countries faced limitations in utilizing their resources due to the lack of technology resources and capacity. As a result, oil power was concentrated on major oil companies commonly referred to as the seven majors (two European and five American oil companies)¹, which had control over production, prices, and competition. They focused on exploring oil reserves in the Middle East and Africa, obtaining control of two-thirds of total global oil production through concessions. Although they offered to pay taxes and royalties to the countries, the concessions mostly favored them and gave these companies exclusive rights to explore for oil and whatever oil they found belonged to them. By 1952, they produced 90% of the crude oil and marketed 75% of the oil product outside North America and the Communist countries (Wilkins, 2011, p.162).

Before 1970, however, the situation shifted as some Middle East countries decided to change the terms of the concessions and nationalize their oil industries, taking control of oil reserves and production from foreign companies, resulting in a transfer of power from multinational corporations to national governments. As a consequence, big oil companies and Western, feeling threatened, organized boycotts; one clear example was when Iran nationalized its oil industry in 1951, which led to the overthrow of Prime Minister Mohammad Mossadegh supported by the U.S. This transfer of power from big oil companies to national governments was also hastened by the rise of independent oil companies which offered more favorable conditions to oil-producing countries.

¹ Anglo-Persian Oil Company (now BP), Royal Dutch Shell, Gulf Oil (Acquired by Chevron in 1985), Standard Oil of California (now Chevron), Standard Oil of New Jersey (now ExxonMobil), Standard Oil of New York – Socony (later became Mobil Oil and then ExxonMobil), Texaco (Acquired by Chevron in 2001)

These events, and as a response to major companies lowering the posted price of oil, motivated the creation of the Organization of the Petroleum Exporting Countries (OPEC) with five of the major oil-producing countries: Saudi Arabia, Kuwait, Iran, Venezuela, and Iraq to improve their negotiating capacity in front of big oil companies and have more control over the production and prices. The creation of OPEC had significant implications for the global oil market. Initially, OPEC's attempts to control prices and use oil as a political weapon proved ineffective, even causing Arab countries to create a parallel organization OAPEC, to impose an embargo on supporters of Israel in any Arab-Israeli conflict. However, during the late 1960s, OPEC expanded its membership to include countries like Abu Dhabi, Algeria, Libya, Nigeria, Indonesia, and Qatar, increasing its share of oil exports to nearly 80%, with Saudi Arabia and Iran alone accounting for almost 25% of the world's total oil production by 1974 (Rose E., 2004).

Moreover, in 1973 when the Middle East conflict erupted the impact of these changes became truly evident since OPEC decided to cut global oil supply by 5% and imposed an embargo on countries such as the United States and the Netherlands that supported Israel, leading to a surge in oil prices and triggering a recession in the Global North countries. These factors along with increasing economic growth, growing oil consumption, and slowed production in the US, which began to import larger amounts of oil to meet their internal demand, led to making OPEC the dominant actor in the global oil market (Rose E., 2004). "By 1968, the majors were producing a little more than 15 percent of the crude oil outside North America and the Communist countries, compared with their 90 percent in 1952" Wilkins, 2011, p.162).

Similarly, the rise of fundamentalist governments contributed to reversing the excessive benefits from majors. For instance, Muammar al-Qadhafi, an Arab nationalist, reached power in 1969 in Libya and ordered a production cut to increase the price of Libyan crude oil, which was capitalized on by other countries. Due to the lack of response from Western countries and oil firms, OPEC pressured again to increase the price in 1970, strengthening OPEC's capacity to dictate oil production and pricing policies to the global community, and "laid the foundation for OPEC to use pricing policies and production cutbacks to generate more revenues for its members" (Rose E, 2004, p.433).

Subsequently, the political dynamics in the Middle East have significantly influenced the power dynamics in the oil market such as the Gulf Wars in 1991 and the Iraq War of 2003 which were partially fought to gain control over oil reserves in Kuwait and Iraq. In 1990, Iraq, led by President Saddam Hussein, invaded Kuwait, claiming it as Iraq's "19th province." This act of aggression was widely condemned, as it posed a threat to regional stability and global oil supplies. In response, an international coalition, led by the U.S. and composed of 34 countries, was formed to liberate Kuwait. Several Arab nations, including Saudi Arabia, Egypt, and Syria, joined the coalition. While in the second Gulf War, initiated by the U.S. was justified by the belief that Iraq possessed weapons of mass destruction and had ties to terrorist organizations. The aftermath left Iraq deeply divided, with ongoing sectarian tensions, political instability, and the rise of extremist groups, and had also consequences for the oil supply and prices increasing ((Rose E., 2004).

At the same time, the rise of non-OPEC producers, particularly the United States and Russia, has reshaped the global energy landscape and decreased OPEC's power. Technological advancements in the oil industry, particularly in the extraction of unconventional oil resources, such as shale oil and tight oil extraction in the U.S. and Canada have played a crucial role in this shift. This development has transformed the United States into one of the world's largest oil producers, reducing its dependence on oil imports and challenging OPEC's traditional dominance. Nevertheless, the emergence of non-OPEC producers has prompted increased

cooperation between OPEC and non-OPEC countries to manage the global oil supply and stabilize prices.

Finally, the current shift in the oil dominance is the emergence of renewable energies which could lead countries heavily dependent on oil revenues to face challenges as the demand for oil declines, potentially impacting their political and economic stability.

III. Environmental and Economic Challenges of Oil

A. Environmental Consequences of Oil Extraction and Consumption

There are many disputes concerning the climate change effect of the oil industry. But there is no doubt that emissions from oil mining and usage have had some effect on global warming. Oil has negative externalities that have not paid exact expenses. Besides the emission effect, oil produces some hazards and has many dangerous aspects when extracted from the earth and the sea and moved to other countries.

According to Seba.T (2015), there have been much more oil leak accidents than those officially announced, causing great harm to the environment, which takes decades to recover, and the cost of conservation is enormous. For instance, BP's Gulf of Mexico oil spill in 2010 affected tens of thousands of square miles. The National Oceanic and Atmospheric Administration's Fisheries Agency banned commercial fishing in June 2010 in 80,000 square miles around the well. The area of BP's spill alone is 80 times the area of solar power plants that can cover the fuel efficiency of all-electric vehicles in the United States.

No	Accident name	Year	Leaking Quantity (Ton)
1	Lakeview gusher oil spring (U.S.)	1910	1,200,000
2	Oil reservoir explosion during the gulf war	1991	900,000
3	Deep water horizon (U.S.)	2010	780,000
4	Eatles ship (Mexico)	1979	530,000
5	Prestage oil ship leaking	2002	250,000
6	Taeen Sea oil leaking (South Korea)	2007	10,900

Source: <https://vbluechip.tistory.com/en8try>

Likewise, despite the international community's carbon reduction efforts, CO2 emissions from oil in 2021 increased by about 13% compared to the early 2000s (Figure 2.). As shown in Figure 3, the demand for oil rises with the growth of energy-intensive economies such as China and India., although the oil era seems to come to an end. Saudi Arabia, the world's largest oil producer, for instance has been promoting a solar power generation project in which it has invested 109 billion dollars since 2011. It is evident that each country is looking for a way out of the oil era, but oil is still the dominant energy resource and origin of power.

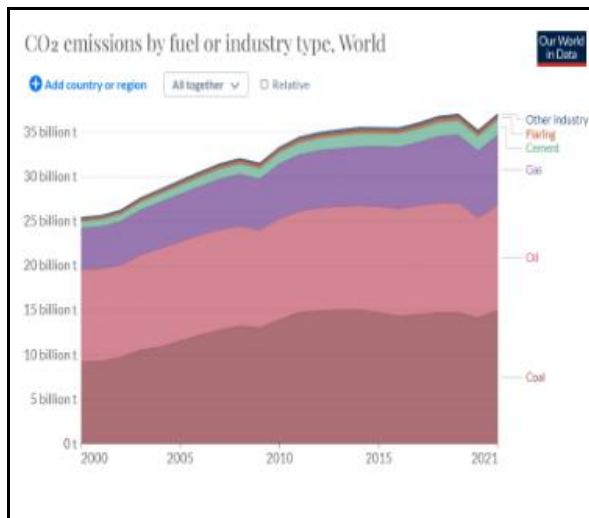


Figure 2. CO2 emissions by fuel
Source: Our World in Data, 2021

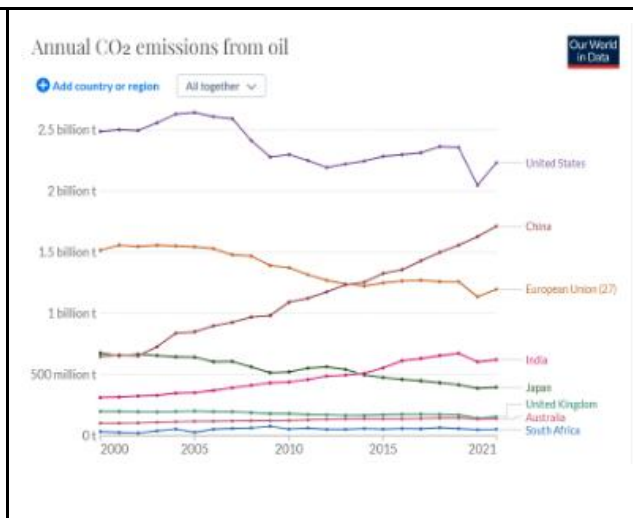


Figure 3. Annual CO2 emissions from oil (each country)
Source: Our World in Data, 2021

B. Economic volatility and Dependence on oil price

Oil is a geopolitical commodity and is very vulnerable to future price forecasts. Compared to other conventional natural resources such as coal, oil production, and price change volatilities are very high. As a result of the war, mentioned, in the Middle East, oil shocks occurred and the world economy, which depended on oil, was severely shocked. In addition, the rise in crude oil prices has created new competitors. Venezuela, Mexico, and Nigeria have increased their crude oil production, and new crude oil from the North Sea and Alaska has been developed. As a result, crude oil prices have continued to fall by about 60% over the past 20 years since the end of the second oil shock in 1980.

Figure 4 shows that the correlations between oil prices and renewable energy investment are strong. As oil prices increase, renewable energy investment rises. Therefore, oil price volatility inevitably leads to renewable energy spread. The geopolitical concentration of oil energy and the resulting price volatility has led to investment in alternative energy. This will be a change not only for all mankind but also for the global environment.



Figure 4. OPEC oil production ratio
Source: Statistics Korea

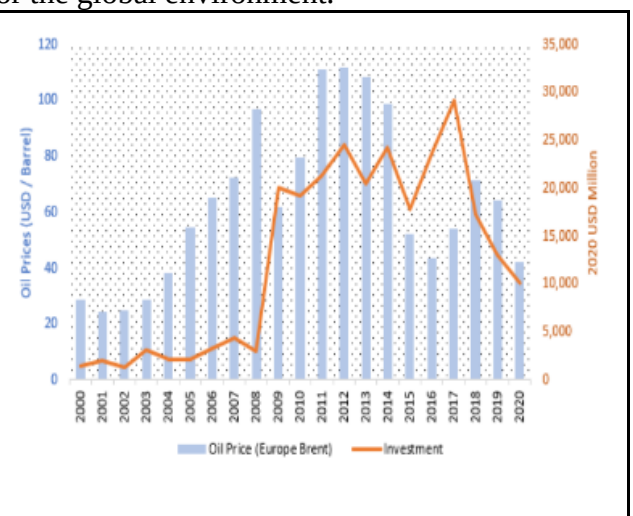


Figure 5. Oil prices vs. renewable energy investment
Source: EIA (US. Energy information administration), IRENA (International renewable energy agency)

According to statistics from Global Data, the global power project scale reached about 5.2 trillion dollars as of last year. By field, wind energy 28%, hydropower 17%, nuclear power 16%, solar power 11%, gas 9%, coal 8%, and others 11%.

IV. Rise of Renewable Energy

A. Overview of renewable energy sources and their potential

Energy Transitioning from fossil to renewable fuels is the primary key to addressing climate change issues (Jang & Gong, 2022). According to Korea Energy Economics Institute, 135 countries around the world declared carbon neutral, 84 set renewable energy targets, 36 of which set renewable energy targets at 100%. The reason why the world focuses on renewable energy is that it has advantages in addition to energy independence. It is noteworthy that renewable energy sources are plentiful and all around us. Sunlight and wind are constantly being replenished and considered environmentally safe compared to traditional fossil fuels. Most importantly, renewable energy has secured price competitiveness. Ten years ago, building a new plant that burns fossil fuels was much cheaper than building a new solar or wind power plant. Max (2020) explained that Wind was 22 percent and solar was 223 percent more expensive than coal. But over the past few years, this has changed completely. renewable sources have a variety of potentials to offer sustainable and clean energy solutions for various applications. The price of electricity from new coal (LCOE) declined by only 2% from \$111 to \$109. While the price of solar decreased from \$378/MWh in 2010 to \$68/MWh in 2019, getting 89% cheaper. Likewise, the price of wind (off and onshore) got 70% cheaper (Figure 5).

Solar Energy:

The source of solar energy is natural sunlight to generate electricity or heat. There are two types of generating electricity: solar photovoltaic (PV) panels and solar thermal power systems. Max (2020) said Electricity from utility-scale solar power is \$359 per MWh in 2009. In just a decade, prices have fallen 89 percent and are relatively cheaper than fossil fuels <figure 7>. Thanks to advanced technology and decreasing costs, solar energy has tremendous potential for large-scale power generation and decentralized applications.

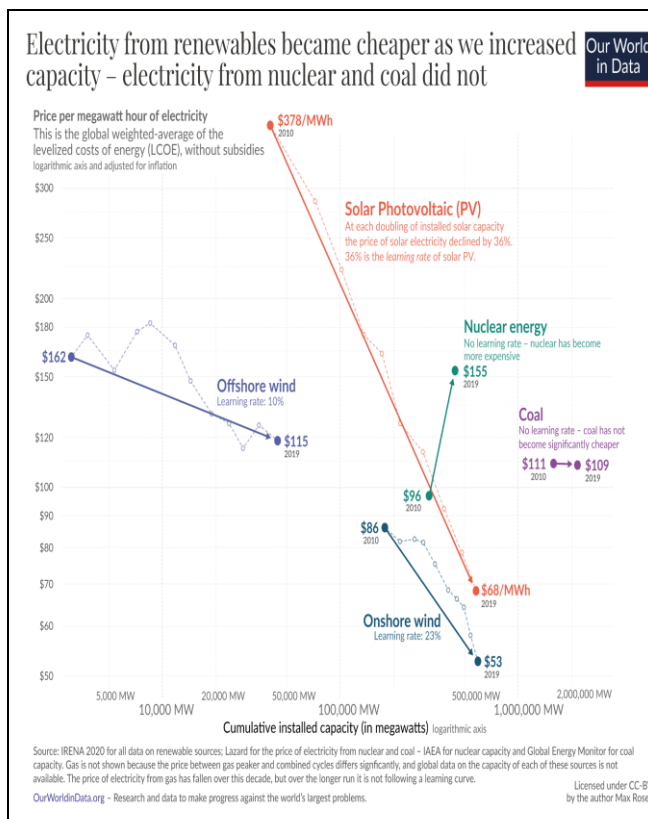


Figure 6. Price changes according to cumulative installed capacity for the last 10 years
Source: IRENA 2020, IAEA for nuclear capacity and Global Energy Monitor for coal capacity.

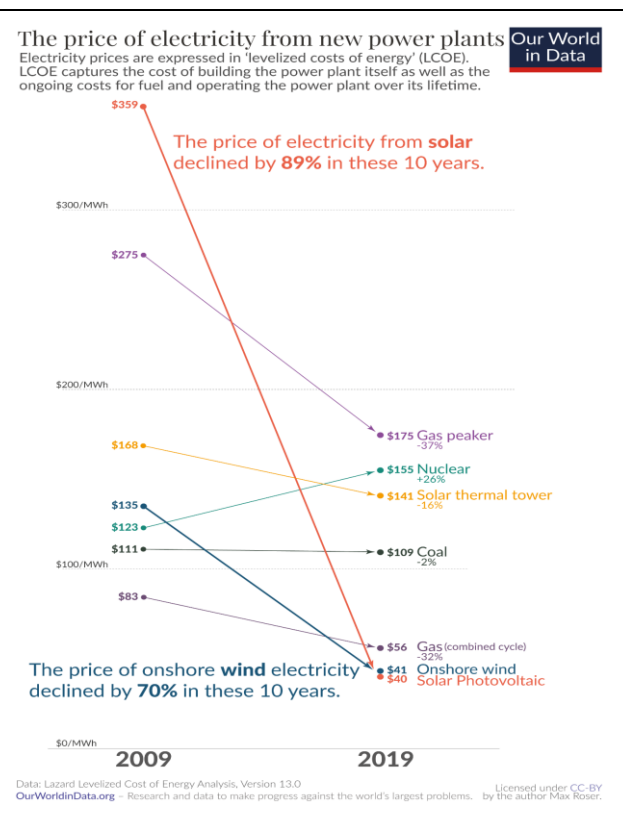


Figure 7. The Price of electricity changes from new power plants for the last 10 years
Source: Lazard Levelized Cost of Energy Analysis, Version 13.0

Wind Energy:

Wind energy is based on the kinetic energy of wind to generate electricity. Wind power generation is the most developed technology and has grown significantly compared to other renewable energy sources. Over the last two decades, wind power plant construction globally has surged and unit production costs have declined as efficiency has been improved. The cost of power production also decreased by more than 70% compared to 10 years ago, securing price competitiveness (Figure 6). Water consumption per unit is the smallest in energy production. Wind turbines do not use water to generate electricity (2030 Future Energy Report, 2012). However, the primary disadvantage is that the wind blows intermittently, so an additional supply from other energy sources is required to meet the base load at the time of peak power demand.

Hydropower:

Hydroelectric energy generates electricity based on the force created by the drop or pressure of water. Consistent technology development of hydropower is possible because there is no waste generated during the power generation process and there is no fear of resource depletion. Compared to other energy sources, power generation production costs are much cheaper and price fluctuations are low, which reduces operating costs. However, the construction of hydroelectric facilities requires sufficient area for dam construction, and there are environmental risks to the ecosystem. Specifically, developing countries such as Congo and Nepal are more dependent on hydroelectric power than developed countries.

Biomass Energy:

Biomass is a method of burning forest, agriculture, and urban waste or obtaining electricity from methane gas emitted from landfills. Biomass is mainly used to power small power plants and cogeneration plants (2030 Future Energy Report, 2012). One of the major benefits of biodiesel is its ability to significantly reduce the emission of smoke from cars. However, its widespread use is hindered by its high cost. This cost is primarily driven by the fluctuation of soybean and other grain prices, which are essential raw materials for biodiesel production. These prices tend to move in line with international crude oil prices, meaning that when oil prices increase, grain prices also rise. In contrast, Brazil has successfully embraced bioenergy, particularly bioethanol, as a popular fuel additive for vehicles. This success can be attributed to Brazil's abundant production of sugar cane, which allows them to take full advantage of bioethanol's benefits, with approximately 70% of their vehicles utilizing this renewable fuel source.

Geothermal Energy:

Geothermal energy is a renewable energy source that harnesses the Earth's natural heat to generate electricity or provide direct heating and cooling. It involves extracting hot water or steam from underground reservoirs and using it to power turbines for electricity generation. Geothermal power plants are typically located in regions with abundant geothermal activity, such as volcanoes, like the United States and Indonesia. The utilization of renewable energy sources holds significant promise in reducing greenhouse gas emissions, mitigating climate change, and enhancing energy security. Their development can play a significant role in transitioning to a sustainable, low-carbon energy transformation. It's crucial to note that the viability and potential of each renewable energy source depend on various factors such as geographical location, generation capacity, economic considerations, and policy support.

B. The rise of renewable energy leaders.

Recently, Covid issues and the Russia-Ukraine war have raised concerns about energy security, leading to an era of increased energy independence. This is linked to reducing dependence on imported petroleum energy and expanding renewable energy that can produce more energy in its own country. Especially, wind and solar are the main sources and are rapidly expanding due to policies that support cost competitiveness and the transition to low-carbon electricity and green hydrogen (BP, 2022). Regarding production costs, the development of technology such as increasing solar module efficiency and turbine loading ratio can make prices more competitive.

Overall, Solar and Wind energy showed a record 12% of electricity generation over the world in 2022, rising slightly by 2% from 10% in 2021 (World Economic Forum). Solar and Wind shares rose by 24% and 17% from the previous year (Malgorzata, 2023). According to IRENA data, the top countries for renewable energy consumption are China, the United States, and Brazil, respectively. China was the top leader in renewable energy installations with a capacity of about 1,207 gigawatts, followed by the U.S. which in second place had a capacity of around 371 gigawatts.

Despite being the world's largest carbon emitter, China was already the giant country with the most solar PV and wind capacity in 2017, producing more than 70% of the global solar PV module supply. According to IHA (The International Hydropower Association), China is the leading producer of hydroelectricity with an installed capacity of 370,160 MW in 2020. In the same year, it also added 71.67 GW of wind power capacity, nearly three times more than 2019's levels. Brazil depends on hydroelectricity for 65% of its electricity. The share of Wind power is 11% and solar power is 2.5 %. Brazil has the greatest potential in wind energy during the

dry season, so it is regarded as a protective measure against the negative impact of low rainfall and the limited distribution of current hydroelectric resources.

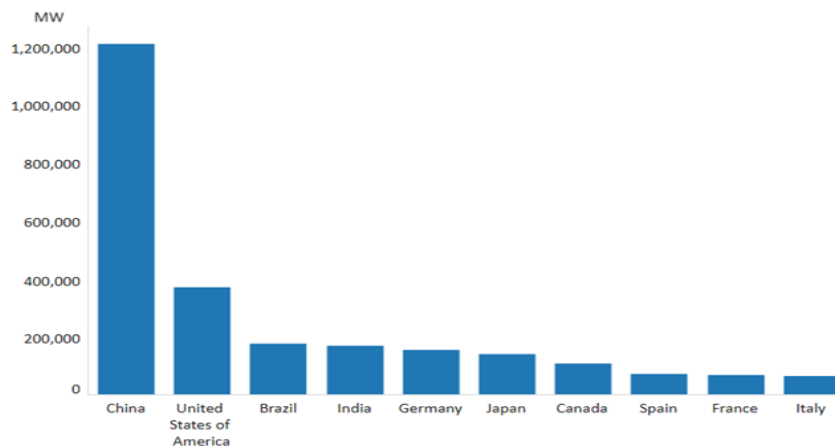


Figure 8. Top Ten Countries' renewable electricity capacity in 2022

Source: IRENA Data, Country rankings (renewable energy)

Additionally, in the case of Denmark in 2021, renewable energy sources contributed to 67% of the electricity supply in the country, with wind energy contributing 46.8% and biomass contributing 11.2%. Denmark aims to become 100% fossil-fuel-free by 2050. Looking into the line graph, the BP Energy Outlook (2023) edition said that the share of fossil fuels in global primary energy will decrease from 80% in 2019 to between 55 and 20% by 2050, while the share of renewable energy is expected to increase from 10% in 2019 to between 35 and 65% by the same year based on the improved cost competitiveness of renewables with supportive policies.

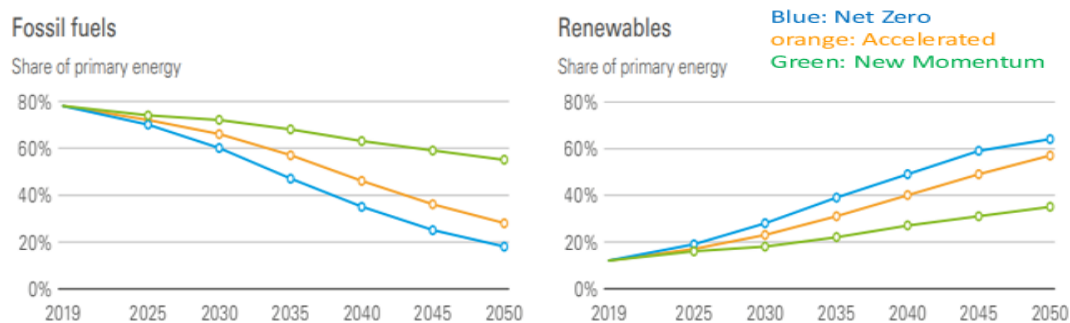


Figure 9. Fossil and renewable share of primary energy depending on three scenarios

Source: BP Energy Outlook 2023 edition

Political power is expected to move significantly in the trend of energy industry change. The political power of traditional fossil fuels will be transferred to renewable energy and storage industries. The power of resource-rich countries and companies will gradually shift to countries and companies with advanced skills and technology.

V. Implications of the Shift from Oil to Renewable Energy

A. Changes in the global energy landscape

The global energy landscape is transforming, driven by the need to address climate change, reduce pollution, and ensure energy security. The transition towards cleaner and more sustainable energy sources is driven by advancements in technology and evolving geopolitical dynamics. The success of various countries in achieving energy transformation largely hinges

on their susceptibility to changes in fossil fuel trade patterns. Equally significant is their position in the competition to dominate the market for green energy technology, with innovation serving as a crucial determinant of the pace of progress.

IRENA (2019) conducted a comparison between selected countries and groups to assess the impact of the energy transition (Figure 9). This comparison examines both fossil fuel exporting-importing countries (the y-axis represents the index of fossil fuel import dependence) and the cumulative number of patents for renewable energy technologies (the x-axis represents the total number of patents held by the respective countries). The findings reveal that countries heavily reliant on imported fossil fuels have positioned themselves at the forefront of the clean energy race, such as China, which has strategically positioned itself as a leader in the clean energy race; the United States boasts strong positions in new technologies through its companies; and Europe and Japan also hold significant positions in renewable technologies and patents. The energy transition is expected to put pressure on the cost of fossil fuels and their profits. In the event of a decline in fossil fuel revenues, these nations will be compelled to reassess their national priorities and strategies.

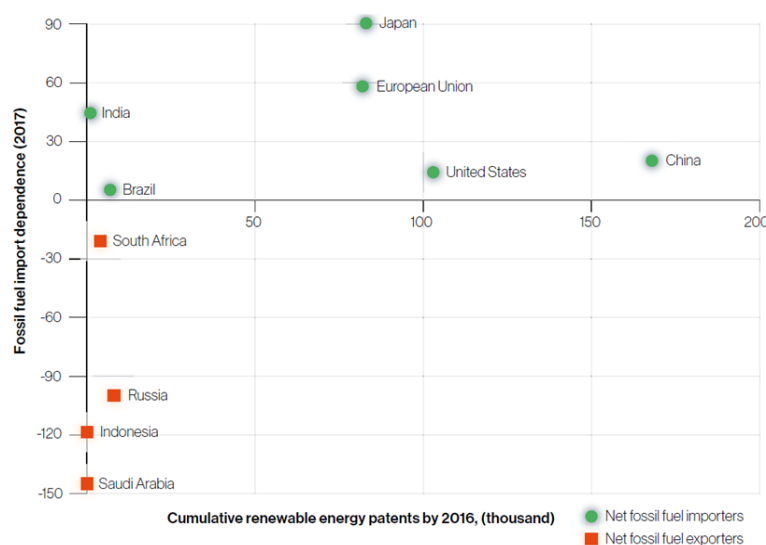


Figure 9. Impact of the energy transition on selected countries and Grouping
Source: BP, IRENA (2019)

B. Measuring the vulnerability of highly dependent economies on oil rents.

Data from the World Bank (2000–2020) reveals that oil rents have consistently accounted for a significantly larger proportion of a country's GDP compared to natural gas rents and coal rents. For instance, countries heavily reliant on fossil fuel rents like Iraq, Kuwait, and Libya have witnessed oil rents comprising 47%, 45%, and 41% of their respective GDPs (Figure 10). Countries experiencing increases in oil rents lead to significant increases in corruption and decreases in political rights (Arezki and Markus, 2009). Consequently, a decline in oil rents could pose substantial challenges for countries that have not adequately prepared their economies for such a scenario. In nations with ineffective governance, the loss of oil revenues could lead to social issues and political instability (IRENA, 2019).

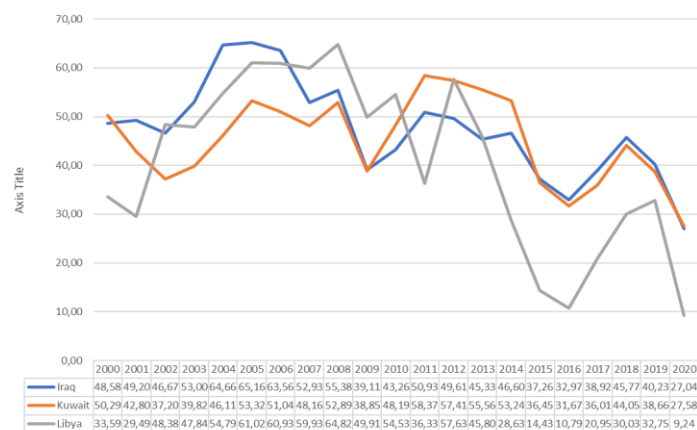


Figure 10. Iraq, Kuwait, and Libya Oil Rents (%) 2000–2020

Source: Processed World Bank Data (2000–2020)

In an attempt to replicate the analysis from the research conducted by IRENA, which evaluates the resilience of the countries to shift in oil rents, comparing fossil fuel rents (including oil, gas, and coal) from 2007 to 2016 about GDP per capita in 2016 (Figure 11), we conducted the same analysis only considering oil rents, the most important source of energy as mentioned before from 2000 to 2020 about GDP per capita in 2020 (Figure 12). The comparison revealed that countries with high oil rents but low GDP per capita are particularly vulnerable to the effects of the energy transition. These countries exhibit characteristics of high exposure (with oil rents comprising more than 20% of their economy) and low resilience (reflected in their limited GDP per capita and financial buffers). Notably, Iraq, Libya, Congo, and Angola are among the countries falling within this group, which aligns with the findings of the IRENA research. This reinforces the argument that oil rents have a significant impact on the overall fossil fuel rents.

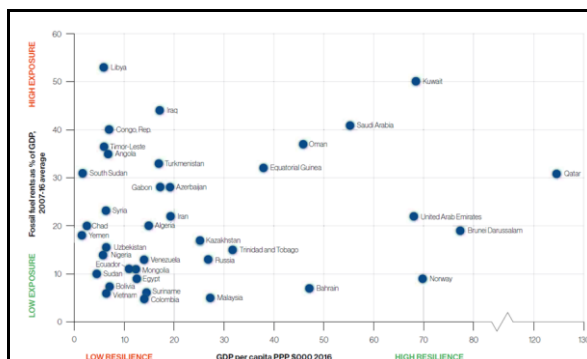


Figure 11. Fossil fuel rents vs GDP per capita, 2007 – 2016

Source: IRENA (2019)

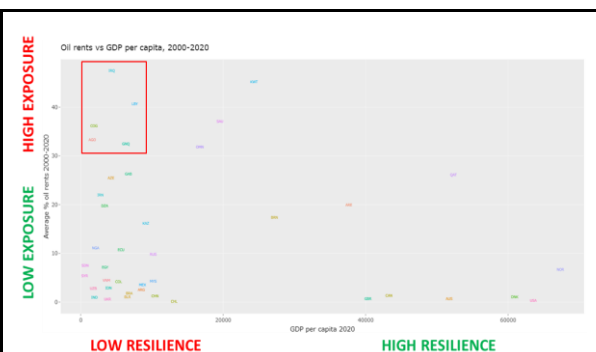


Figure 12. Oil rents vs GDP per capita, 2000 – 2020

Source: Processed World Bank Data (2000–2020)

C. Measuring power from oil rents

With the aim to analyze the political power held by oil-producing countries and the possible shift occurring with the emergence of renewable energies, we conducted a simple correlation analysis. To assess political power, we use the Gross Domestic Product (GDP), which according to Beckley (2018), is the most widely used indicator that reflects the value of all goods and services produced within a country over a specific period, capturing both economic and military capacities, as well as through this a country can buy influence through aid, loans, investment, and bribes and to cultivate soft power. Also, we use oil rents as an indicator, considering their significant impact on both fossil fuel rents and the overall economy. Correlation analysis was employed to determine the correlation between oil rents and GDP as

a starting point. A high correlation should indicate a high dependency of the economy on oil rents but also may show a possible proxy of an oil exporting country and an important influence in the oil market. The analysis was run for 43 countries whose oil reserves information was available and prior to applying the correlation analysis, in order to normalize the data, for better interpretation, we applied a logarithmic transformation to both variables. To illustrate the country's progress over time, we divided the period into decades. The graphical representation of the correlation analysis provides several key insights:

1. Saudi Arabia has consistently maintained high oil rents in line with a stable GDP. On the other hand, the United States experienced high oil rents from 1970 to 2000, but starting in 2001, there was a decline in oil rents, and Russia emerged as a prominent player in the global oil market.
2. China has been able to catch up with the United States economically without making significant changes in oil rents. This indicates that oil rents do not play a crucial role in driving China's economic growth.
3. There is no significant impact of having large oil rents on the economic side for several countries, such as Gabon, Congo, and Equatorial Guinea. As previously mentioned, increases in oil rents often result in significant increases in corruption and a decrease in political rights, which can hinder economic growth.
4. Countries have shown a tendency to increase their oil rents over the decades, as evidenced by the fact that countries with low oil rents in the 1970-1990 period have since increased their oil rents through extensive exploration.



Figure 13. Analysis of Oil Political Power
Source: Processed World Bank Data (1970-2020)

It is worth mentioning that this is a simple correlation analysis that must be complemented with other variables such as oil reserves, for example, to accurately evaluate countries' influence and power in the oil market. Also consider that political power based on oil also can be measured through information about the control over oil infrastructure, the location of oil-producing countries or regions, or the level of international alliances.

Economic diversification is a crucial factor in reducing dependence on oil rents, and it has long been a prominent policy objective for numerous oil-exporting countries (IRENA, 2019). To achieve a secure pathway toward economic transformation and diversification, a customized strategy is essential. Policy interventions aimed at promoting diversification

encompass fiscal and monetary discipline, investments in human capital and infrastructure, as well as support for the private sector.

D. Impact on geopolitical alliances and rivalries:

Countries that are capable of harnessing new renewable energy technologies can anticipate an expansion of their global influence and reach. In our analysis, we explore the potential for a shift in global political power resulting from the adoption of renewable energy. We argue that countries with substantial renewable energy capacity possess the potential to emerge as leaders in renewable energy and thereby wield political power and influence. To assess the level of renewable energy influence or political power, we use renewable energy capacity as a key metric. Additionally, we employ GDP as a measure of overall political power (Figure 14).

We again applied a correlation analysis for over 200 countries to examine the relationship between renewable energy capacity (electricity capacity) and GDP. To enhance data visibility, we applied a logarithmic transformation to both variables. In order to capture the country's progress over time, we divided the period into two decades: 2000-2010 and 2011-2020. Although limited by the available data, the graph offers several significant findings:

1. During the first decade (2000-2010), China and the USA occupied similar positions in terms of renewable energy capacity. However, countries known for their oil resources, such as Kuwait, Qatar, and Libya, were not present in the renewable energy sector as they had not yet ventured into its development.
2. In the second decade (2011-2020), China significantly expanded its renewable energy capacity and emerged as a leader in this field. Meanwhile, oil-producing countries began to shift their national plans and diversify their economies by incorporating renewable energy as a new energy source. This indicates a strategic move towards reducing dependence on fossil fuels and embracing sustainable alternatives.
3. An example of such a shift can be seen in the case of Qatar, which decided to leave OPEC in 2018 and redirect its focus toward natural gas exports. Qatar has set a target to generate 20% of its electricity from renewable energy sources and has specifically prioritized solar and waste-to-energy as key areas of development. This illustrates Qatar's commitment to diversifying its energy portfolio and embracing renewable sources as a strategic choice.

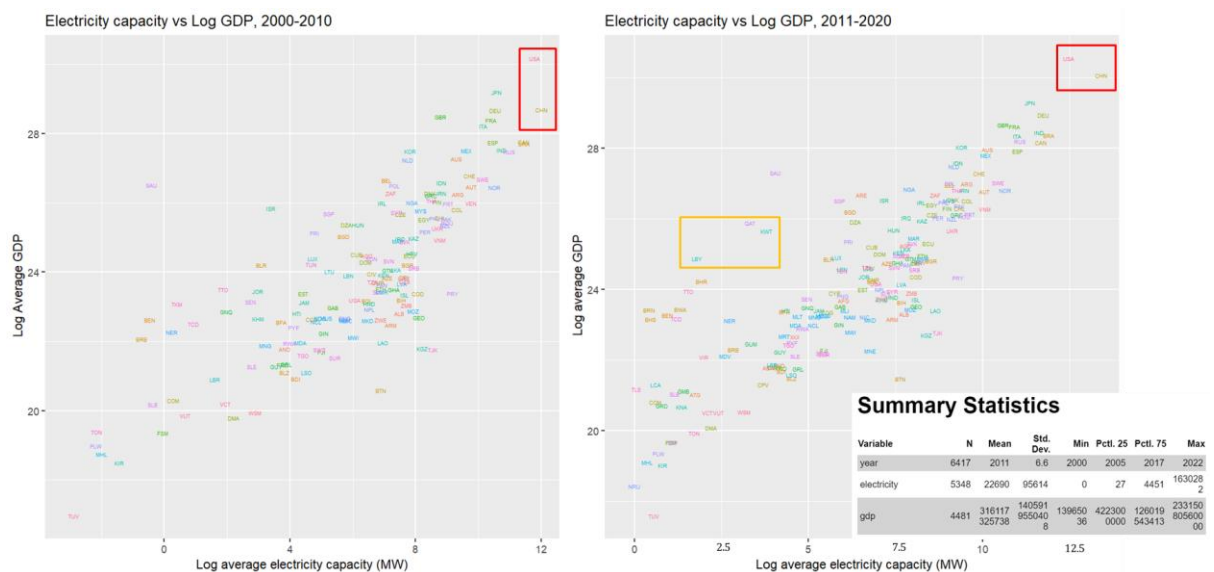


Figure 14. Analysis of Renewable Energy Political Power
Source: Processed IRENA and World Bank Data (2000-2020)

There are three impactful actions that countries can take to become global leaders in renewable energy:

a. Influence in shifting alliances:

Renewable energy will not only have an impact on the balance of power between countries but also reshape alliances, and trade flows, and establish new interdependencies centered around electric grids and emerging commodities. In the event of a decline in global demand for fossil fuels, alliances formed based on fossil fuels are likely to weaken, causing the energy pillar to become relatively less significant for countries. An example from OPEC highlights the potential reinvention of the organization, yet it is expected to diminish in significance within a world characterized by a growing array of energy alternatives. Qatar's recent decision to withdraw from OPEC, shifting its focus towards natural gas exports, exemplifies the challenges faced by the organization amidst rapid and structural changes in the market (Hubbard and Stanley, 2018).

Numerous new alliances and initiatives are emerging to promote multilateral cooperation and propel specific renewable technologies forward. One noteworthy example is the International Solar Alliance, which originated from the COP 21 Paris Climate Conference in 2015, with India playing a pivotal role as the initiator (UNFCCC, 2015). While these alliances are currently in the early stages of development and primarily concentrate on technological cooperation, they are expected to have a significant geopolitical impact in the future.

b. Shifting to geographic trade:

Renewable energy sources will give rise to new patterns of connections and interdependencies between countries and regions. In general, the dependence on global energy markets will shift towards regional grids. Countries that presently rely on oil imports from distant regions will endeavor to establish their renewable energy capabilities and foster integration with neighboring nations' electricity grids. This approach will position electricity as a key component of their energy strategies. Consequently, there is an anticipated shift towards a more regional focus in global energy markets. Nonetheless, advancements in ultra-high-voltage transmission technologies, which reduce energy losses during long-distance transmission, hold the potential to enable electricity trade over larger distances. The control and management of grid infrastructure will become increasingly critical in terms of ensuring national security and exerting global influence and power.

Paraguay serves as a notable example in the realm of renewable energy. It boasts the highest percentage of renewable energy per capita globally and stands as the largest net electricity exporter in South America, with 85% of its generated electricity being exported. Furthermore, Paraguay is among the world's major net electricity exporters (Pappis et al., 2021). The country relies primarily on hydropower (67%) and biomass (33%) for its domestic primary energy supply (IEA, 2017). As of 2020, Paraguay's electricity was sourced entirely from hydroelectric power, predominantly from two large projects: the Yacyreta dam (3.2 GW) and the Itaipu dam (14 GW). Nevertheless, despite this achievement, 75% of the total electricity produced (55,282 GWh) from these plants is exported to Brazil and Argentina (IEA, 2021).

c. Shifting in foreign energy policy:

The energy transition may lead to a decrease in the geostrategic importance of oil and gas as tools for foreign policy. There have been cases where countries have intervened in oil and gas markets to pursue their foreign policy goals. Japan serves as an example of a country that has shifted its focus from exclusively securing fossil fuel imports to actively including renewables, with a particular emphasis on hydrogen. In December 2020, Japan unveiled its "Green Growth Strategy in line with Carbon Neutrality in 2050." The government has set forth an ambitious plan that entails expanding renewable energy

sources, revitalizing nuclear power, and deploying new technologies such as low-carbon hydrogen, advanced nuclear reactors with enhanced safety measures, and carbon recycling to decarbonize the electricity sector. According to the strategy, it is envisioned that renewables will contribute to approximately 50%-60% of the electricity demand in 2050, with nuclear and thermal power plants equipped with carbon capture utilization and storage (CCUS) accounting for 30%-40%, and hydrogen and ammonia generation making up 10% (IEA, 2021).

VI. Conclusion

The global energy landscape is currently undergoing a transformative shift towards cleaner and more sustainable energy sources. In this transition, countries that have traditionally depended heavily on exporting and importing fossil fuels are now taking the lead in the race toward clean energy. These countries have positioned themselves at the forefront due to their ownership of patents for innovative technologies in the renewable energy sector. On the other hand, countries that have high oil rents but low GDP per capita face particular vulnerability to the effects of the energy transition. These nations are in a precarious position due to their high exposure to the risks associated with the transition, coupled with limited resilience in terms of financial buffers and resources.

Our analysis of oil rents and GDP reveals several key findings. Firstly, countries have demonstrated a tendency to increase their oil rents over the decades, with some nations successfully maintaining high oil rents in line with a stable GDP. However, this pattern is not consistent across all countries. Some nations have managed to achieve economic growth without a significant increase in oil political power, while others have large oil rents that do not translate into substantial economic growth. In conclusion, we can assert that oil can indeed have a significant impact on an economy and wield influence, particularly in countries with effective governance and the ability to leverage their oil resources strategically.

Our analysis of renewable energy capacity and GDP reveals compelling findings. China has demonstrated significant growth in its renewable energy capacity, positioning itself as a leader in this field while experiencing remarkable economic growth. This progress brings it closer to the United States, which has been a prominent player in renewable energy. Simultaneously, oil-producing countries have begun to undergo a shift in their national plans, diversifying their economies by integrating renewable energy as a new and alternative energy source. This transformation signifies the substantial impact of the energy transition on oil-rich nations. In conclusion, the transition to renewable energy has had a profound impact on oil-producing countries, prompting them to diversify their economies and invest in renewable energy and somehow keep for themselves the power in the global energy market. Moreover, countries with significant renewable energy capacity have the potential to emerge as leaders in this field, thereby exerting political power and influence.

Additionally, there are several impactful actions that countries can take to become global leaders in renewable energy. These include influencing shifting alliances, adapting geographic trade patterns, and adjusting foreign energy policies. International cooperation and supportive policy are needed in order to diversify the energy sources and make it accessible for all countries, as well as to overcome challenges associated with renewable energies such as availability (24/7) and expensive technologies to implement this kind of energy but also for storage it.

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